

In numerical analysis, the Crank–Nicolson method is a finite difference method used for numerically solving the heat equation and similar partial differential equations. It is a second-order method in time. It is implicit in time, can be written as an implicit Runge–Kutta method, and it is numerically stable.

Q. Solve by Crank–Nicolson Method $\frac{\partial^2 u}{\partial x^2} - \frac{\partial u}{\partial t} = 0$ $0 \leq x \leq 1$
 Subjected to the condition $u(0,t) = 0$, $u(1,t) = 0$, $u(x,0) = 100x(1-x)$
 Taking $h = 0.25$ find one step

Solution:

$$\begin{aligned} \text{① } u(0,t) &= 0 & x=0, \quad t \geq 0 \\ \text{② } u(1,t) &= 0 & x=1, \quad t \geq 0 \\ \text{③ } u(x,0) &= 100x(1-x) & x=x, \quad t=100x(1-x) \end{aligned}$$

By CNM: $\frac{\partial^2 u}{\partial x^2} = a \frac{\partial u}{\partial t}$ $k = ah^2$

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} - \frac{\partial u}{\partial t} &= 0 \\ \frac{\partial^2 u}{\partial x^2} &= \frac{\partial u}{\partial t} \quad a=1 \\ k &= 1 \times 0.25^2 \\ \boxed{k} &= \boxed{0.0625} \end{aligned}$$

→ h

$x \backslash h$	0	0.25	0.5	0.75	1
0					
0.06					

↓
K

→ h

$x \backslash h$	0	0.25	0.5	0.75	1
0	0	18.75	25	18.75	0
0.06	0				0

↓
K

$$y = x, t = 100x(1-x)$$

$$0, t = 0$$

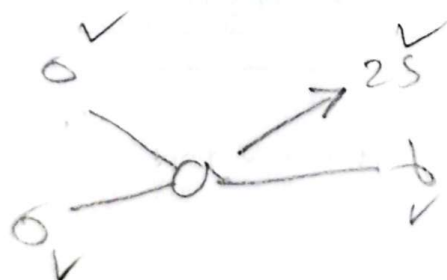
$$x = 0.25 \quad t = 100 \times 0.25(1-0.25) = 18.75$$

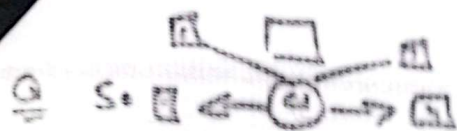
$$x = 0.5 \quad t = 100 \times 0.5(1-0.5) = 25$$

$$x = 0.75 \quad t = 100 \times 0.75(1-0.75) = 18.75$$

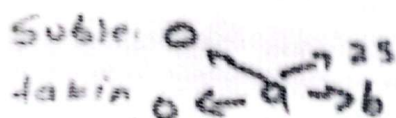
→ h

$x \backslash h$	0	0.25	0.5	0.75	1
0	0	18.75	25	18.75	0
		a	b	c	

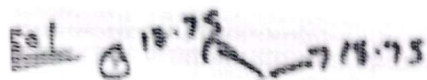




$$a = \frac{1}{4} [1 + 2 + 3 + 4]$$



$$a = \frac{1}{4} [25 + b] \quad \dots (1)$$



$$b = \frac{1}{4} [a + 18.75 + 14.75 + c] \quad \dots (2)$$



$$c = \frac{1}{4} [b + 25] \quad \dots (3)$$



$$(1) \rightarrow 4a = 25 + b$$

$$= 4a - b = 25 \quad \dots (1)$$

$$(2) \rightarrow 4b = a + 37.5 + c$$

$$4b - a - c = 37.5$$

$$-a + 4b - c = 37.5 \quad \dots (2)$$

$$(3) \rightarrow 4c = b + 25$$

$$4c - b = 25$$

$$c - b + 4c = 25 \quad \dots (3)$$

$$a = 9.82$$

$$b = 14.28$$

$$c = 9.82$$

→ h

	$\bar{x}$	0	0.25	0.5	0.75	1
↓	0	0	18.75	25	18.75	0
			a	b	c	
k	0.06	0	9.52	14.28	9.52	0